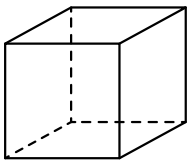
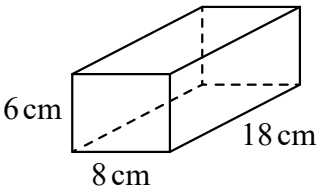
9 The diagram shows a cube and a cuboid.





The total surface area of the cube is equal to the total surface area of the cuboid.

Janet says,

"The volume of the cube is equal to the volume of the cuboid."

Is Janet correct?
You must show how you get your answer.

Cuboid surface area = $2(6\times8 + 6\times18 + 8\times18)$

$= 2(48 + 108 + 144) = 600\ cm^2$.

Cube: $6s^2 = 600$, so $s^2 = 100$ and $s = 10$ cm.

Cube volume = $10^3 = 1000\ cm^3$.

Cuboid volume = $6 \times 8 \times 18 = 864\ cm^3$.

No: equal surface areas do not give equal volumes.

5 marks total
Three face areas
Surface area 600
Cube side 10
Both volumes
No: 1000 ≠ 864

(Total for Question 9 is 5 marks)